

Aquachill Systems (I) Pvt. Ltd.

Client	CEAT, Halol	Doc No ACS-P-416-F-01-018
Consultant	Innowell, Chennai	Rev. No - R0
Project	Bunbury	Date- 02.06.2025

Technical Datasheet of Insulation, Acoustic**Thermal insulation****THERMAL INSULATION (NITRILE RUBBER) :**

Supply of Thermal insulation made out of Class-O closed cell elastomeric nitrile rubber insulation, density of 40-70 kg/m3, operating temperature range shall be -50 °C to +110 °C, thermal conductivity of 0.037 W/m.*K at 20 °C, water vapour resistance as per BS EN12086 (DIN 52615), moisture resistance factor ≥ 7000 , water vapour permeability not more than 0.024 perm/inch, fire performance Class-O complying to BS 476 Part 6 & 7, UL 94, self extinguishing, anti drip, fungi resistance as per ASTM G 21, UL-181, E 2180, zero ODP & GWP, dust and fibre free, CFC/HCFC free, VOC/SVOC free, low VOC water based self adhesive, joints sealed with 50 mm wide aluminum foil tape with all necessary supports and accessories. If insulation is not self adhesive type, separate adhesives used shall be of similar make of nitrile rubber insulation. VOC content of adhesive shall not exceed 850 grams/litre. Mode of measurement of thermal insulation shall be based on surface area.

Sr.No	Description	BOQ Specification	Vender to specify
1	Type	Thermal	Comply
2	Material	Closed cell Nitrile rubber	Comply
3	Class of insulation	Class O	Comply
4	Thermal Conductivity 'K' value, not Exceed	0.037 W/m.*K @ 20 °C	Comply
5	Density	40 - 70 kg/m3	Comply
6	Water Vapour Permeability	NMT 0.024 perm/inch	Comply
7	Adhesive	Low VOC of pedlite or equivalent	Comply
8	13 mm Thik	920 Sqmt	Noted
9	19 mm Thik with glass cloth	350 Sqmt	Noted

Acoustic insulation**ACOUSTIC INSULATION (NITRILE RUBBER) :**

Supply of Acoustic insulation made out of Class-1 open cell elastomeric nitrile rubber, density of 140-220 Kg/m3, minimum 0.55 NRC as per ISO-354 & ASTM-C 423, complying to BS 476 part 7, UL 94, thermal conductivity of 0.047 W/m.K at 20 °C, fibre free and antimicrobial complying to ASTM G21 and ASTM 2180, low VOC water based self adhesive, operating temperature range shall be -20 °C to 85 °C, zero ODP & GWP, joints sealed with 50 mm wide black glass cloth tape with all necessary supports and accessories. If insulation is not self adhesive type, separate adhesives used shall be of similar make of nitrile rubber insulation. Acoustic insulation shall be lined internal to the ducts. VOC content of adhesive shall not exceed 850 grams/litre. Mode of measurement of acoustic insulation shall be based on surface area.

Sr.No	Description	BOQ Specification	Vender to specify
1	Type	Acoustic	Comply
2	Material	Open cell Nitrile rubber	Comply
3	Thickness of insulation	25 mm	Comply
4	Class of insulation	Class 1	Comply
5	Thermal Conductivity 'K' value, not Exceed	0.047 W/m.K at 20 °C	Comply
5	Density	140-220 kg/m3	Comply
6	Adhesive	Low VOC of pedlite or equivalent	Comply
	Qty	15 Sqmt	Noted

Pipe Insulation

Insulation on pipe shall be 9mm thick Plain Nitrile rubber insulation.

Sr.No	Description	BOQ Specification	Vender to specify
1	Type	Thermal	Comply
2	Material	Closed cell Nitrile rubber	Comply
3	Class of insulation	Class O	Comply
4	Thermal Conductivity 'K' value, not Exceed	0.035 W/mK @ 0°C	Comply
5	Density	40 - 60 kg/m3	Comply
6	Finish	-	
7	Adhesive	Low VOC of pedlite or equivalent	Comply
8	Thickness of Insulation	13 mm	Comply
1	25 NB	45	Noted

